LLMs Where are we now?

Abu Bakr Soliman

https://linkedin.com/in/bakrianoo

LLMS EVERYWHERE

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NOVEMBER 30, 2022

ChatGPT

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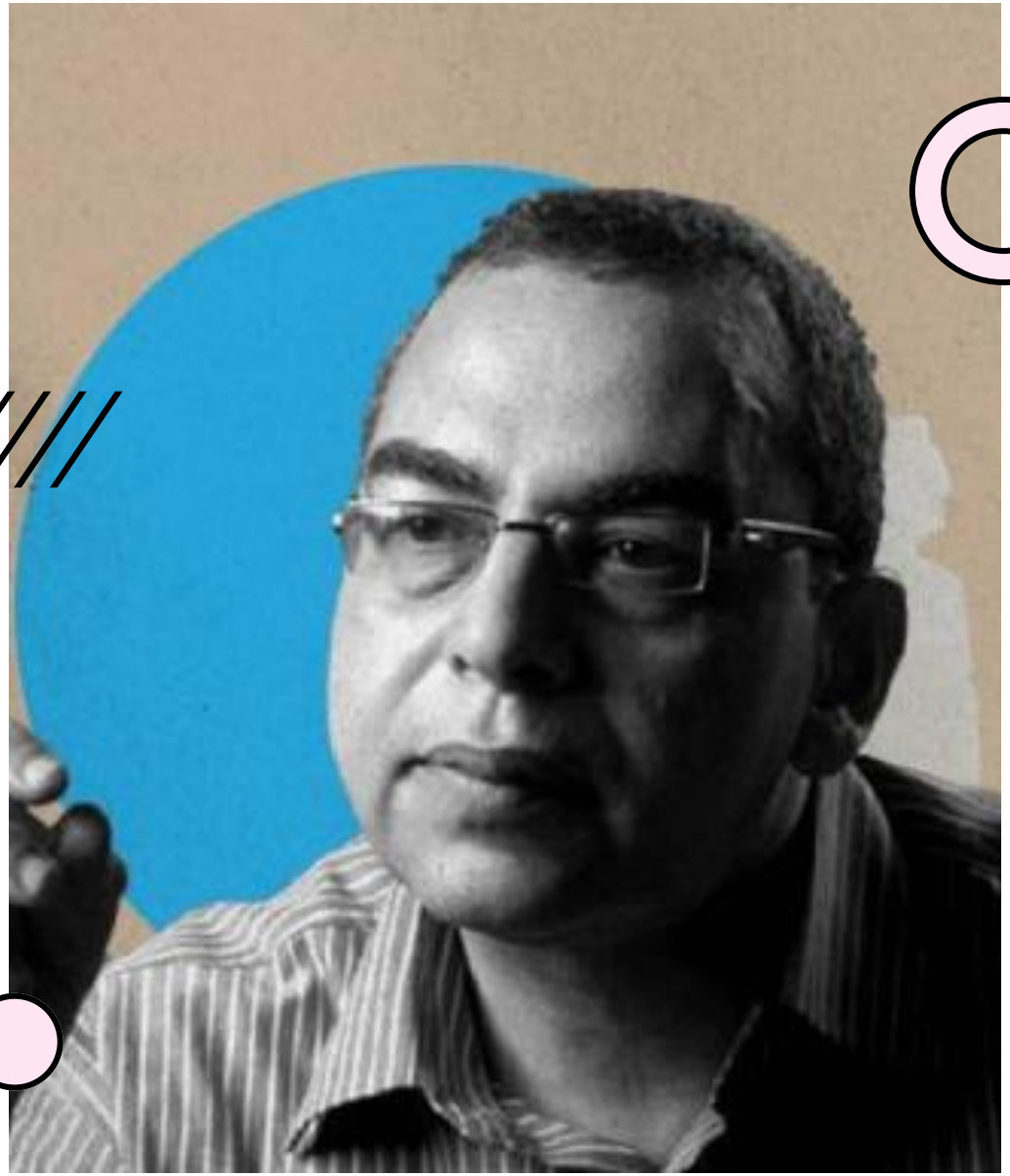
HOW CAN WE TELL THE STORY?



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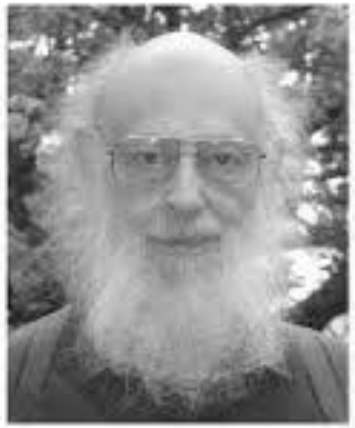
1956 Dartmouth Conference: The Founding Fathers of AI

John MacCarthy

Marvin Minsky

Claude Shannon



Ray Solomonoff

Alan Newell

Herbert Simon

Arthur Samuel

Oliver Selfridge

Nathaniel Rochester

Trenchard More

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NEURAL NETWORKS

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- Can you guess the value?

$$4 * X = 10.4$$

Can you guess the value?

$$4 * X = 10.4$$

#	X`	Predicted	Actual	LOSS	Up/Down

Can you guess the value?

Loss = +29.6

4 * 10 = 40 -> 10.4

#	X`	Predicted	Actual	LOSS	Up/Down
1	10	4 * 10 = 40	10.4	+29.6	D

Can you guess the value?

Loss = +9.6

4 * 5 = 20 -> 10.4

#	X`	Predicted	Actual	LOSS	Up/Down
1	10	4 * 10 = 40	10.4	+29.6	D
2	5	4 * 5 = 20	10.4	+9.6	D

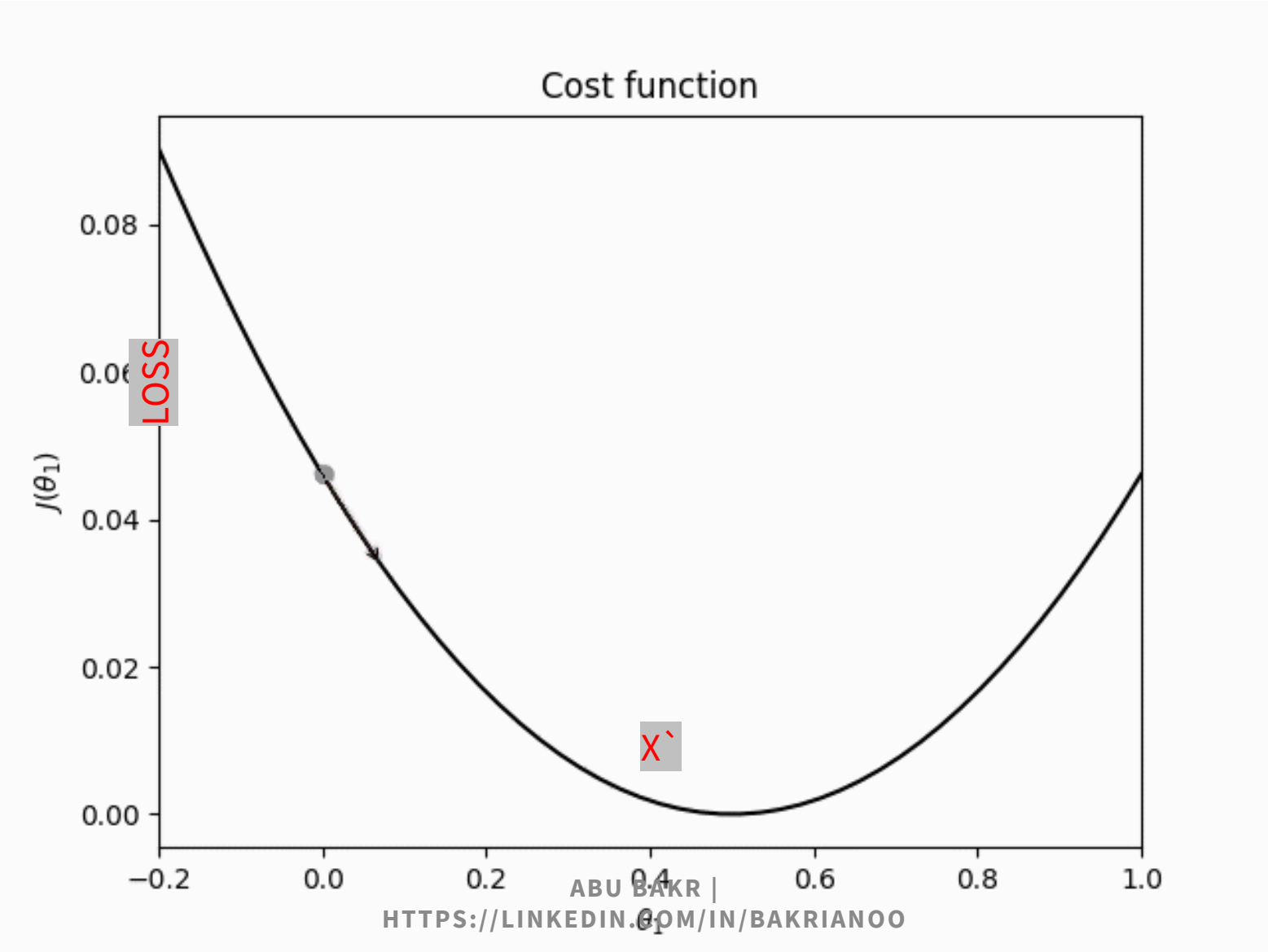
Can you guess the value?

Loss = -0.4

4 * 2.5 = 10 -> 10.4

#	X`	Predicted	Actual	LOSS	Up/Down
1	10	4 * 10 = 40	10.4	+29.6	D
2	5	4 * 5 = 20	10.4	+9.6	D
3	2	4 * 2 = 8	10.4	-2.4	U
4	3	4 * 3 = 12	10.4	+1.6	D
5	2.5	4 * 2.5 = 10	10.4	-0.4	U
	..	..	…	..	..

Gradient Descent



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coursera

Machine Learning Specialization

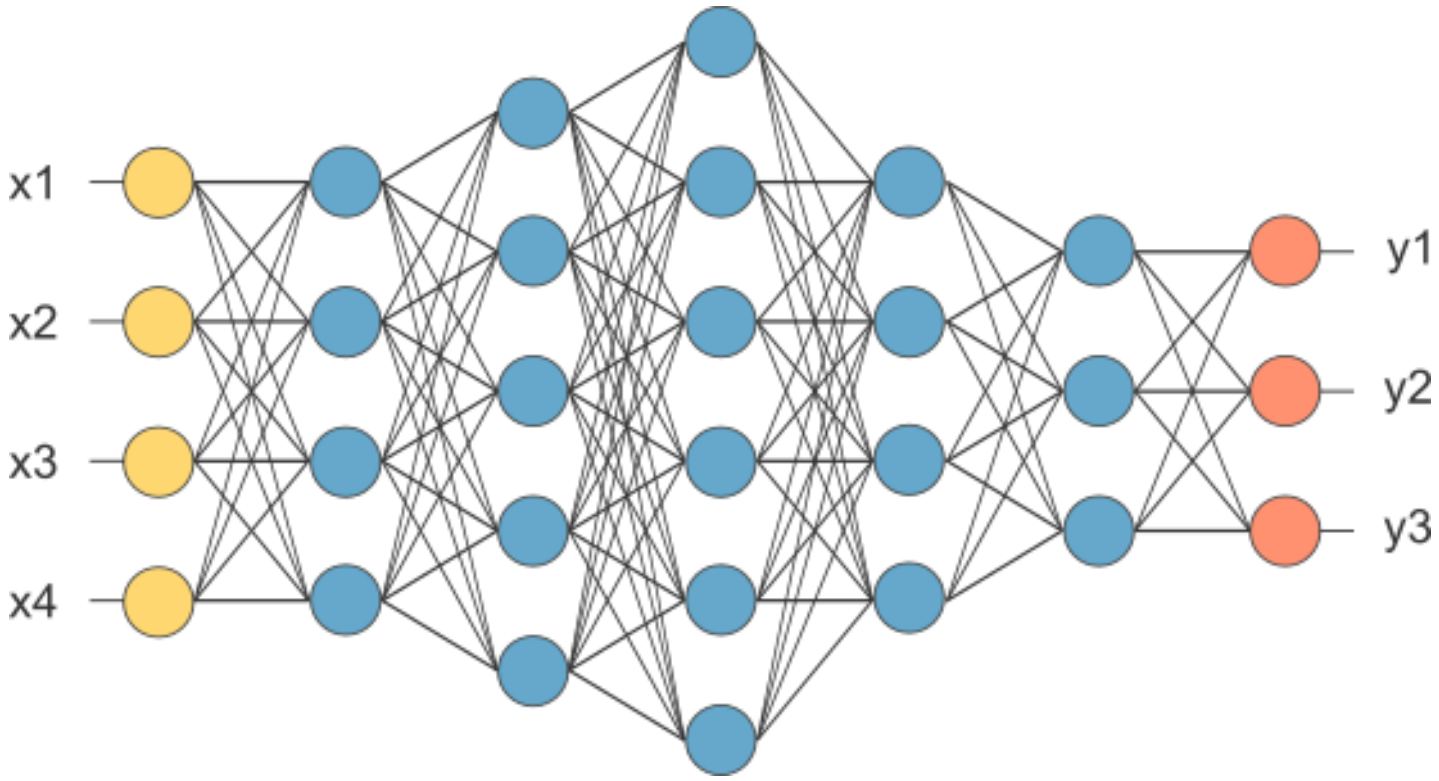
DeepLearning.AI Stanford ONLINE

DeepLearning.AI

Deep Learning Specialization

CERTIFICATE OF COMPLETION

How does it work?



https://www.tutorialspoint.com/artificial_neural_network/artificial_neural_network_supervised_learning.htm

TRANSFORMERS

THE POWER OF
UNSUPERVISED DATA

Google
BERT

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Masked language models

Input Sentence
تتميز سيناء بطبيعة خلابة في جميع فصول السنة
لو سمحت ممكن تقولي فين اقرب مطعم
...
خبر عاجل : الحياة أصبحت أحلى بحمد الله

Masked Language Model

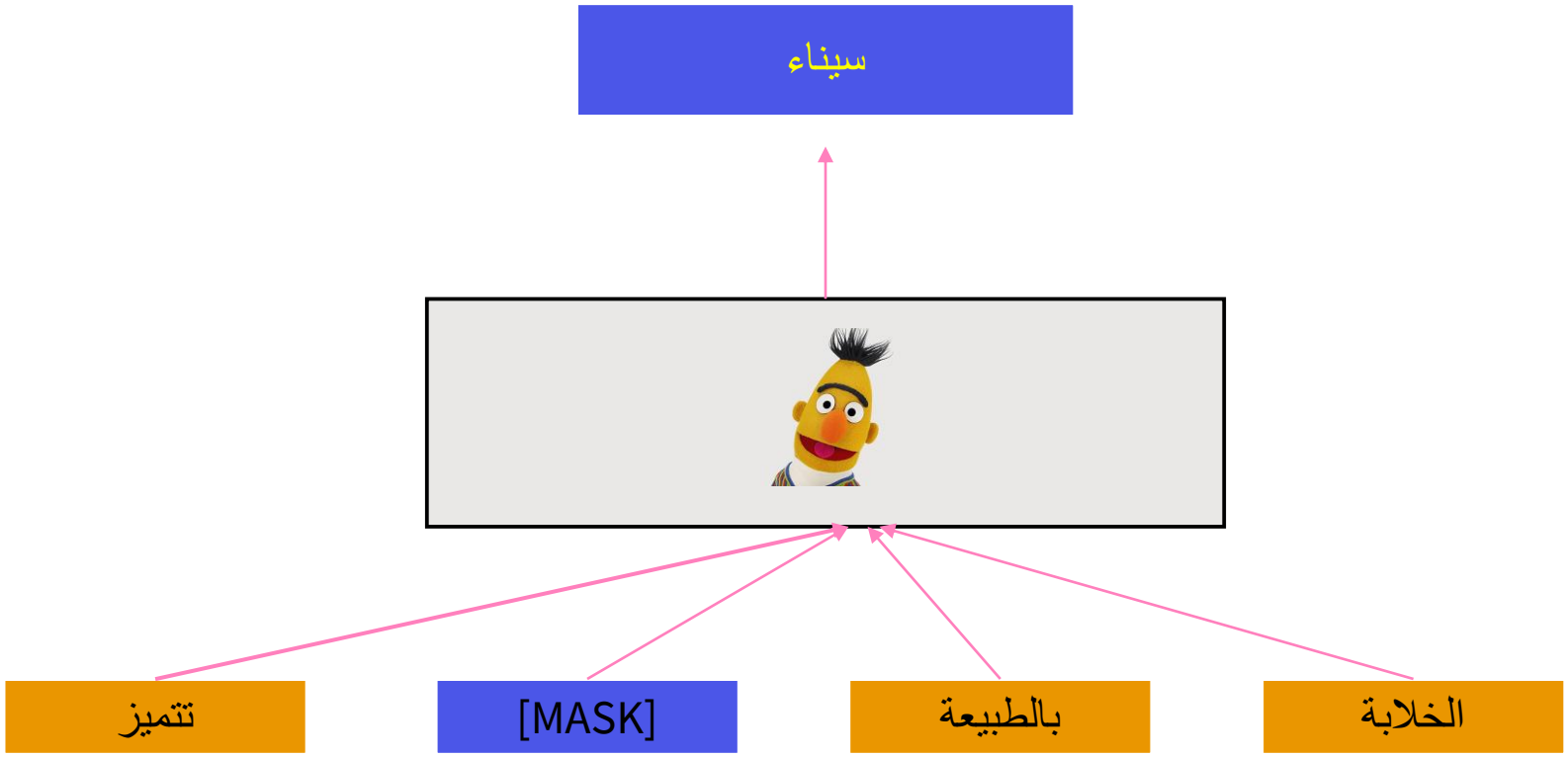
Input	Output
تتميز <MASK> بطبيعة خلابة في جميع فصول السنة	سيناء
لو سمحت ممكن <MASK> فين اقرب مطعم	تقولي
...	
خبر عاجل : الحياة أصبحت <MASK> بحمد الله	أحلى

Predict the next word

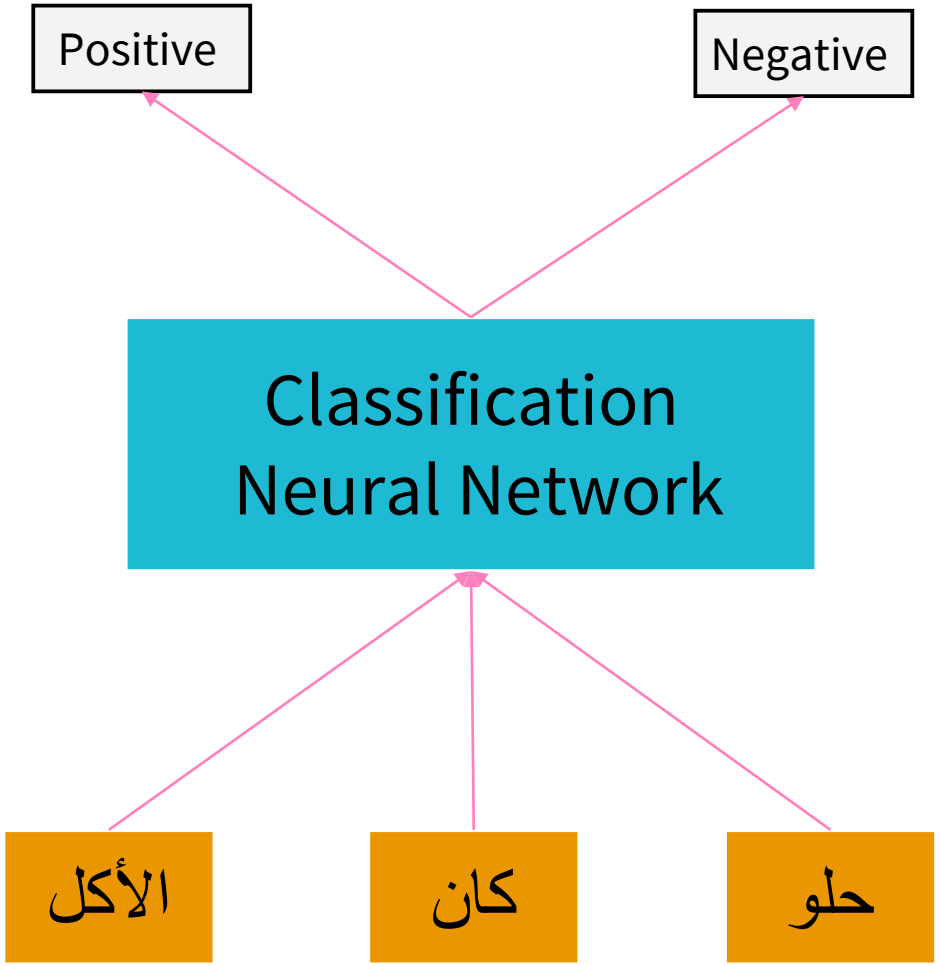
Once upon a time there was a dear little girl who was loved by every one who looked at her

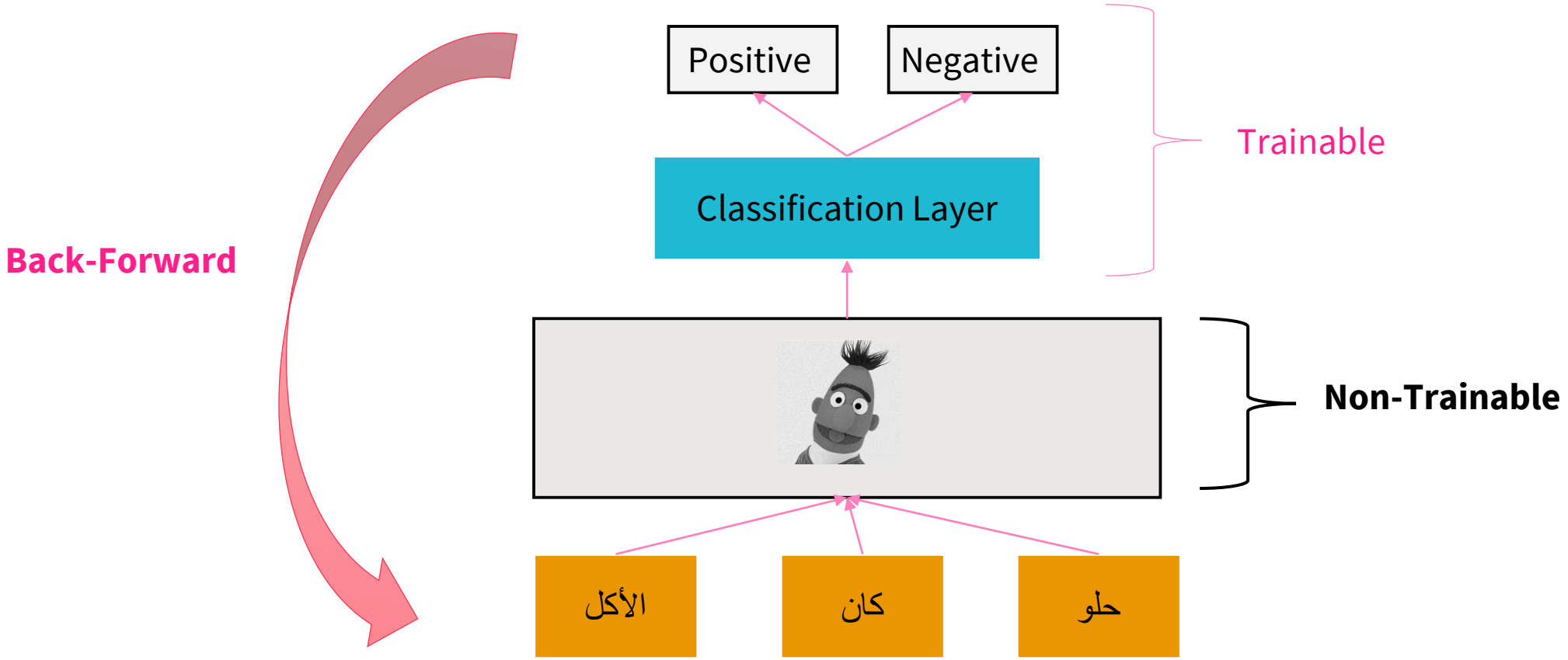
Raw Data

Input	Output
Once	upon
Once upon	a
Once upon a	time
Once upon a time	there
..	..
..	..

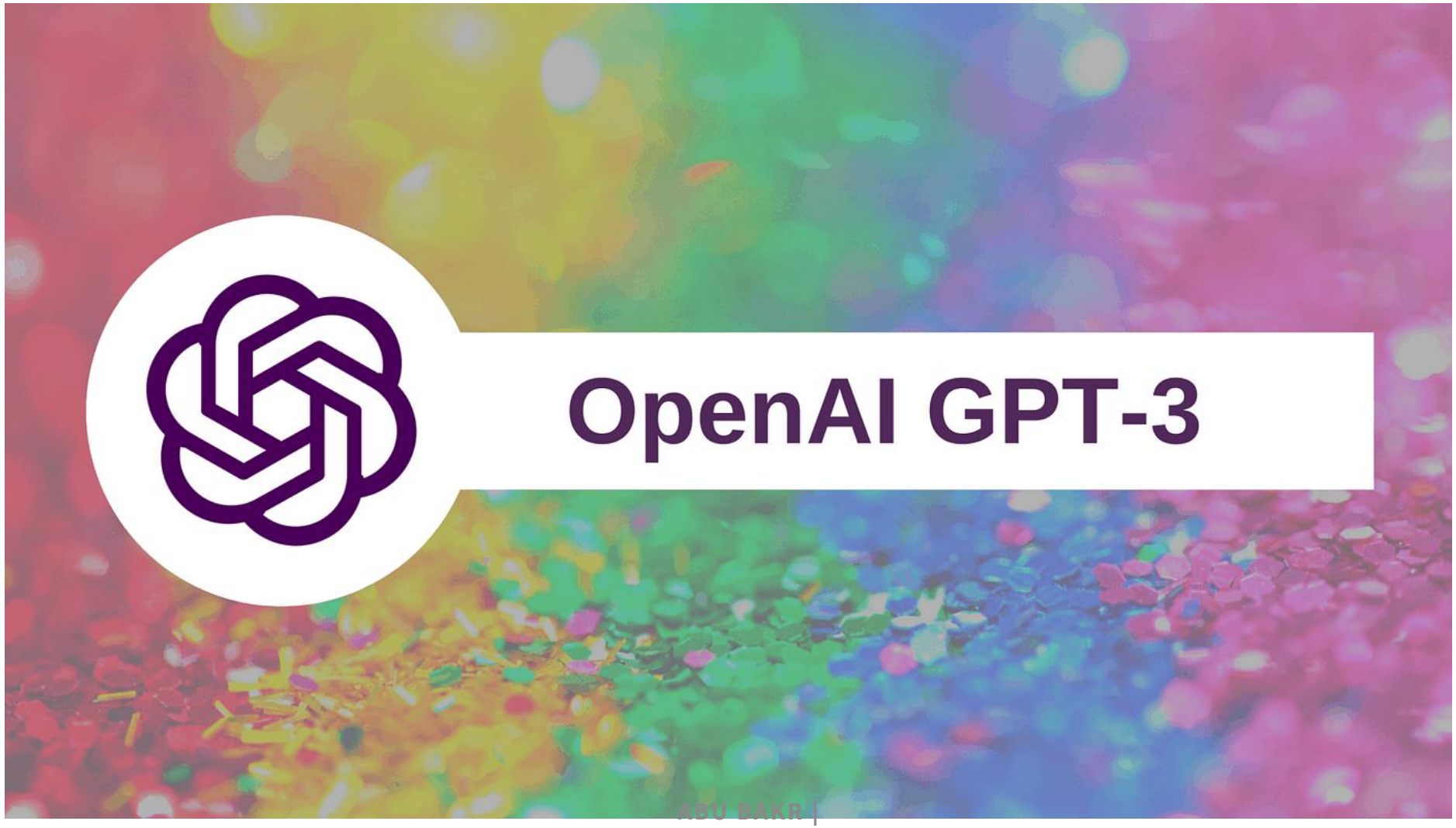

سيناء
تتميز
[MASK]
بالطبيعة
الخلابة

Classic Training

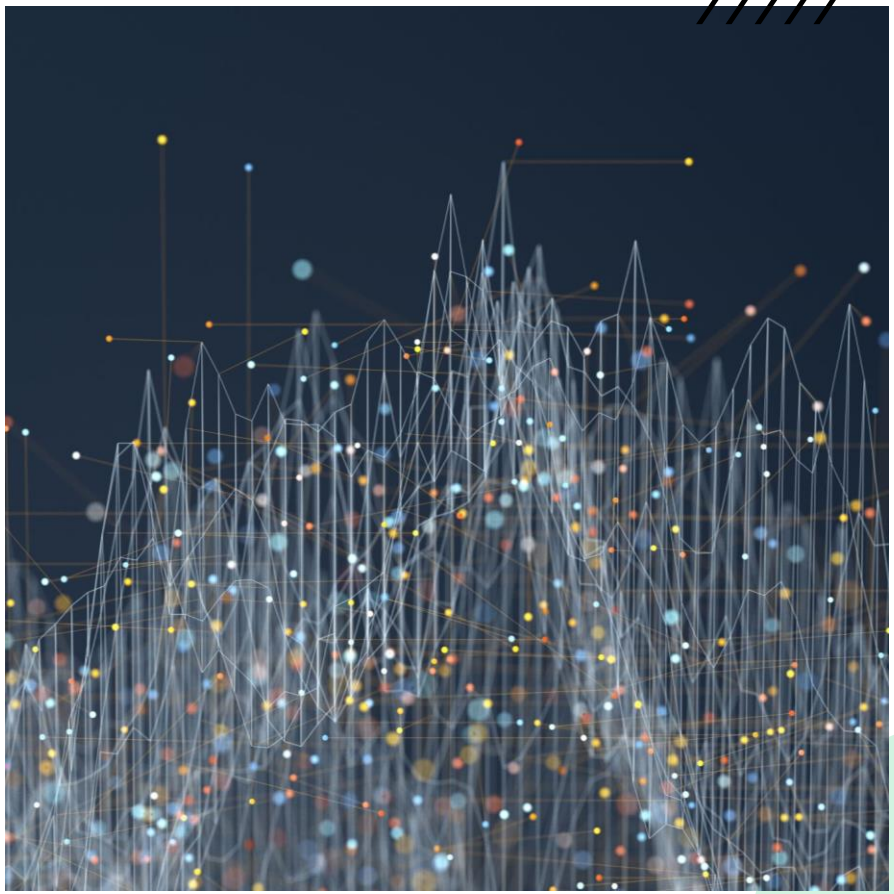



Positive
Negative
Trainable
Classification Layer
Back-Forward
Non-Trainable
الأكل
كان
حلو

June 11, 2020



HOW TO COOK A LARGE LANGUAGE MODEL?



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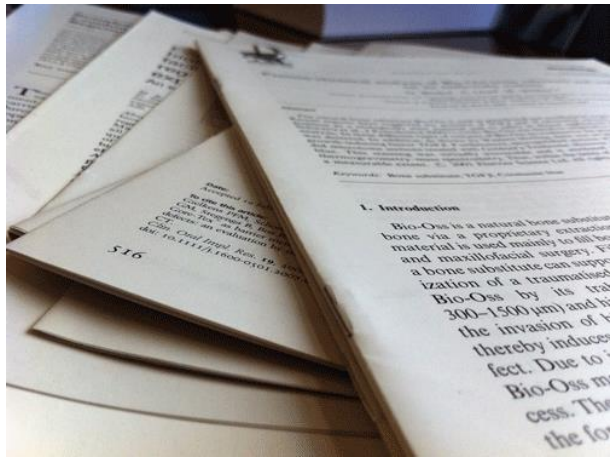
1) Large Data Corpus

COMMON CRAWL

2) Knowledge

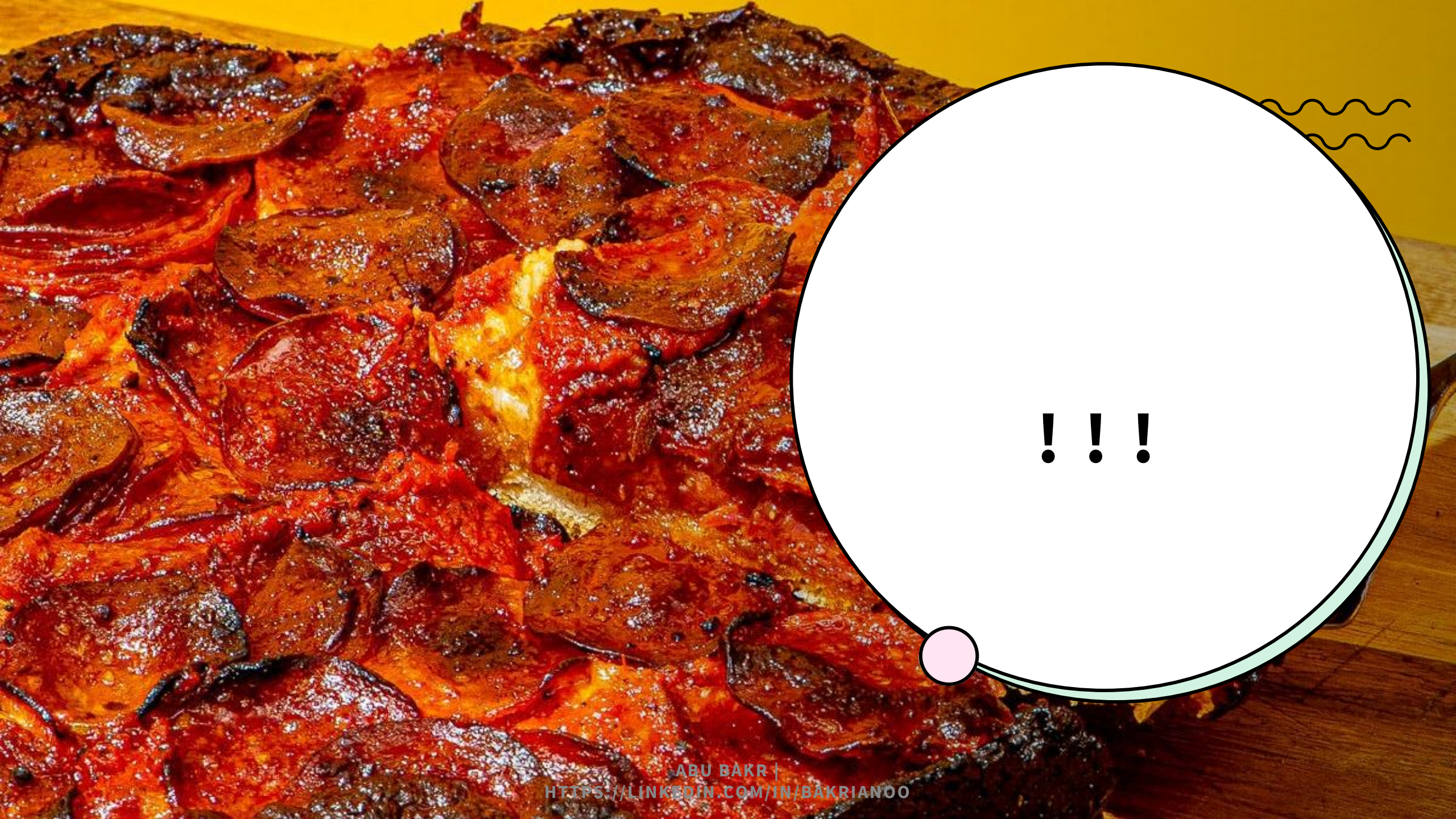


+ PyTorch

3) Resources

+

Data + Codes + Hardware + Waiting …



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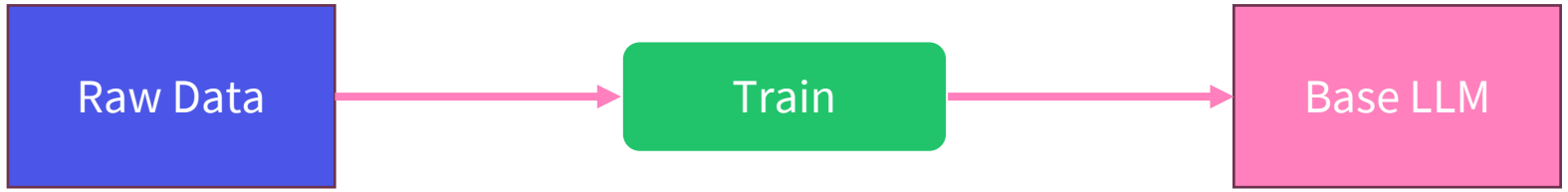
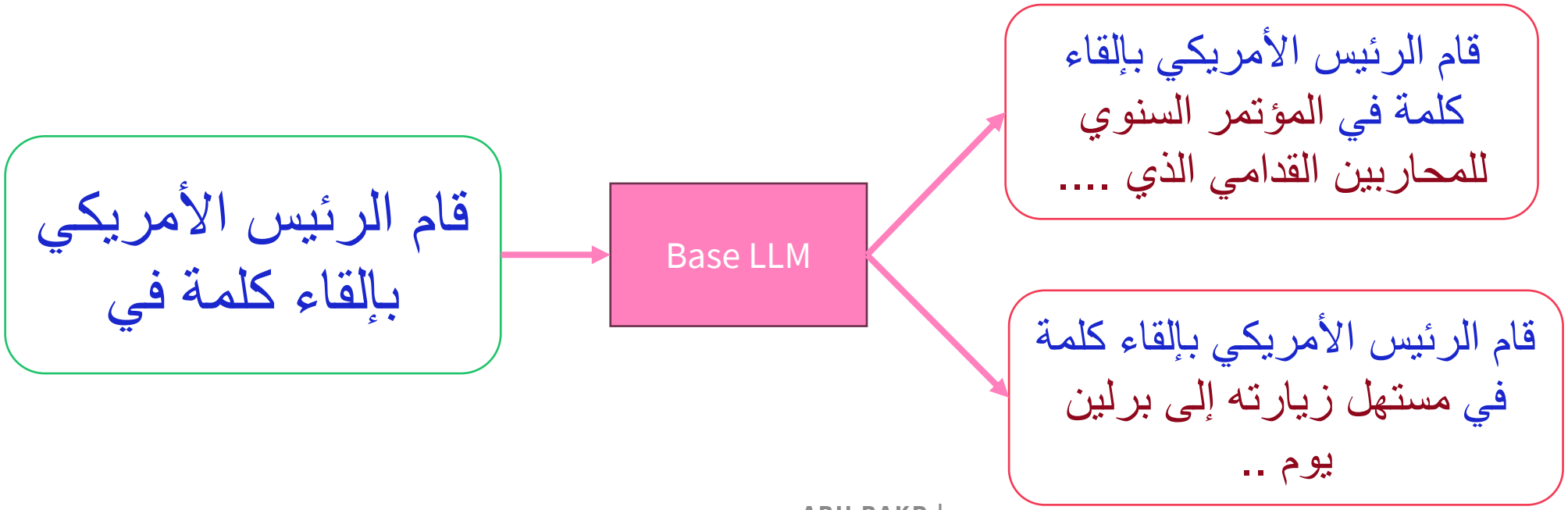
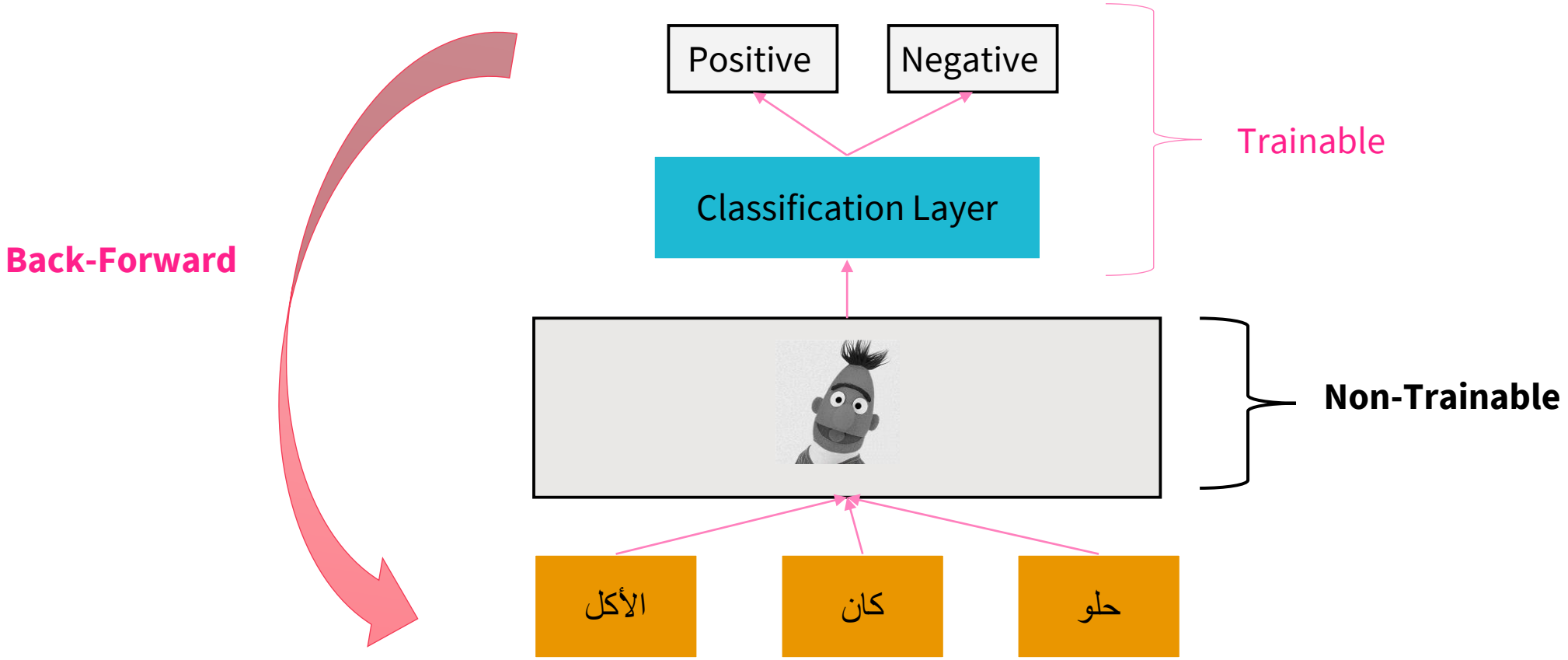
Base LLM


Raw Data
Train
Base LLM


قام الرئيس الأمريكي بإلقاء كلمة في
Base LLM
قام الرئيس الأمريكي بإلقاء كلمة في المؤتمر السنوي للمحاربين القدامي الذي ....
قام الرئيس الأمريكي بإلقاء كلمة في مستهل زيارته إلى برلين يوم ..

Remember This !!



Large Language Model (LLMs)

- Supervised Learning – Follow the instruction

Input (Task)	Output (Answer)
How deep is the Atlantic Ocean?	8,486 m

Large Language Model (LLMs)

- Supervised Learning – Follow the instruction

Input (Task)	Output (Answer)
How deep is the Atlantic Ocean?	8,486 m
Can you flip the letters of the word ocean?	naeco

Large Language Model (LLMs)

- Supervised Learning – Follow the instruction

Input (Task)	Output (Answer)
How deep is the Atlantic Ocean?	8,486 m
Can you flip the letters of the word ocean?	naeco
You have five oranges, how many oranges do you need for a total of eight	Three oranges

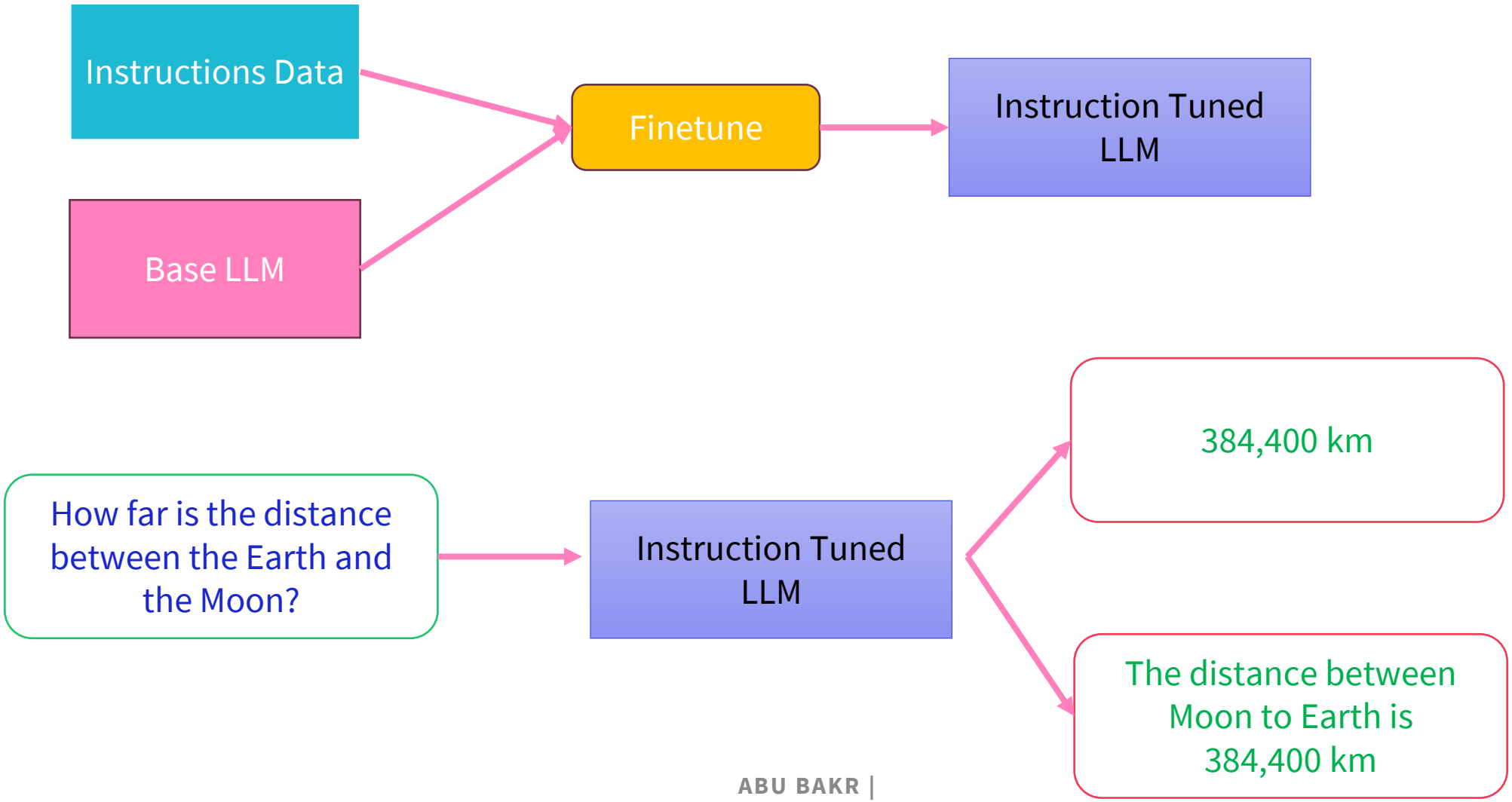
Large Language Model (LLMs)

- Supervised Learning – Follow the instruction

Instructions Data

Input (Task)	Output (Answer)
How deep is the Atlantic Ocean?	8,486 m
Can you flip the letters of the word ocean?	naeco
You have five oranges, how many oranges do you need for a total of eight	Three oranges
..	..
Write me an email to apologize for work today	Dear Manger, I will not be able ..

Instruction Tuned LLM



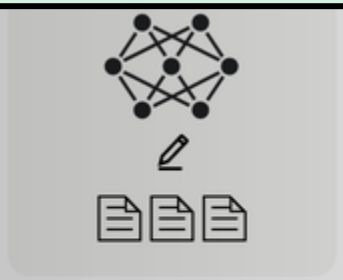
Step 1

Collect demonstration data and train a supervised policy.

A pro
samp
prom

A labe
demo
desire
behav

This data is used to fine-tune GPT-3.5 with supervised learning.

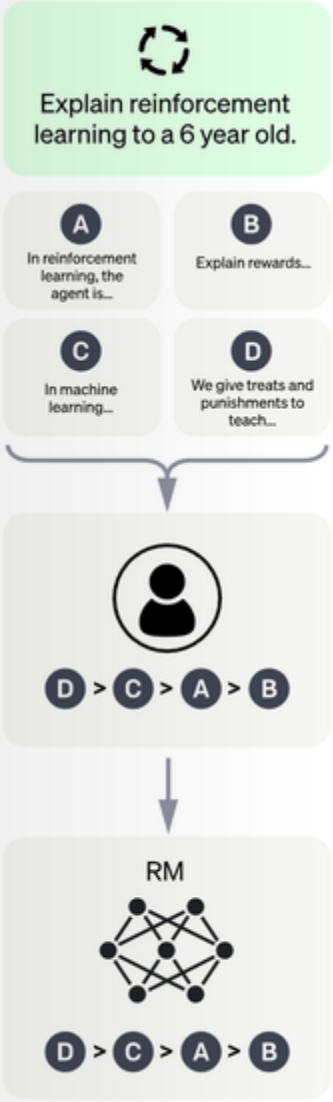


CHATGPT

Step 2

Collect comparison data and train a reward model.

This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

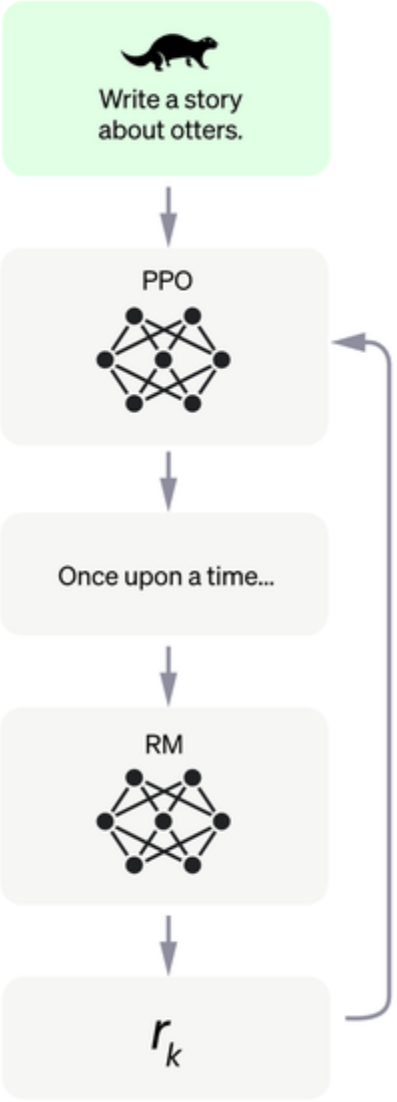
A new prompt is sampled from the dataset.

The PPO model is initialized from the supervised policy.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



ChatGPT

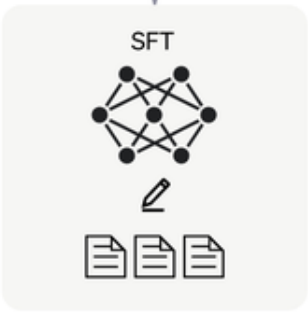
Step 1

Collect demonstration data and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3.5 with supervised learning.



ChatGPT

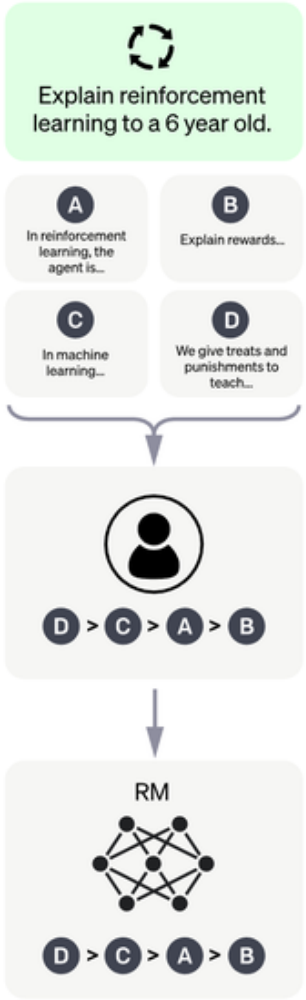
Step 2

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.

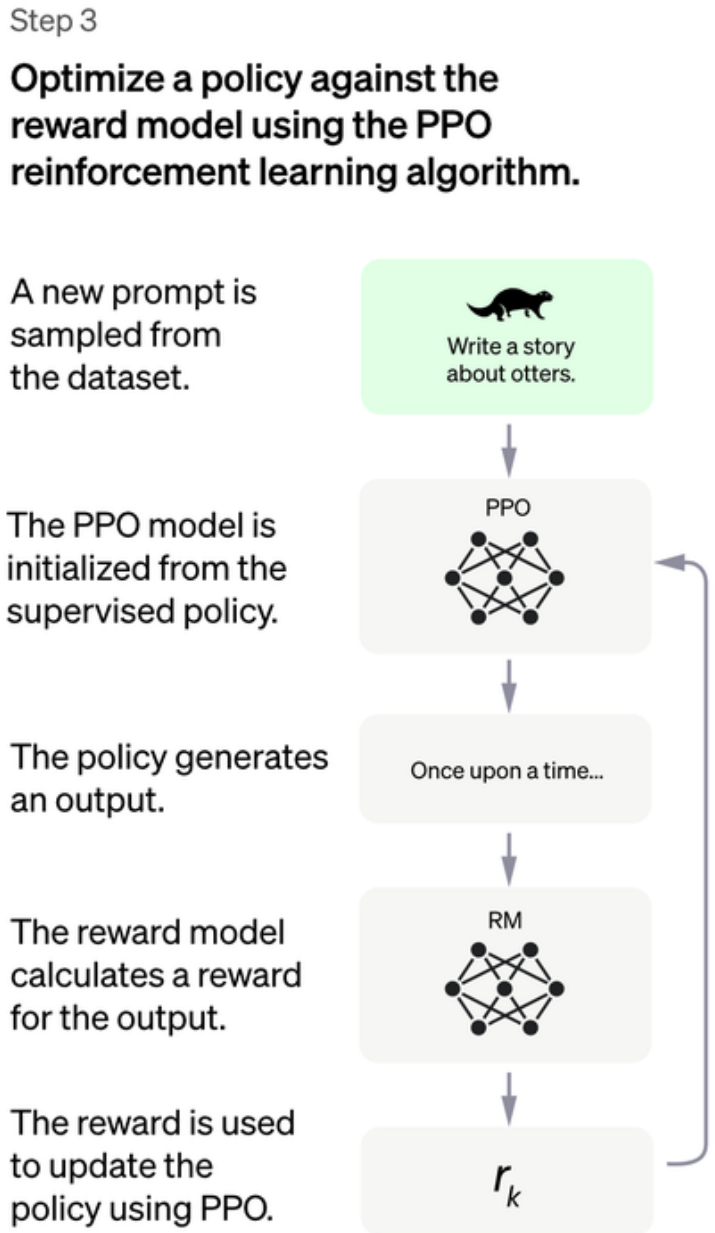
A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



ChatGPT

Reinforcement learning from human feedback (RLFH)



GANGS OF OPEN SOURCE

LLama

LLaMA 1

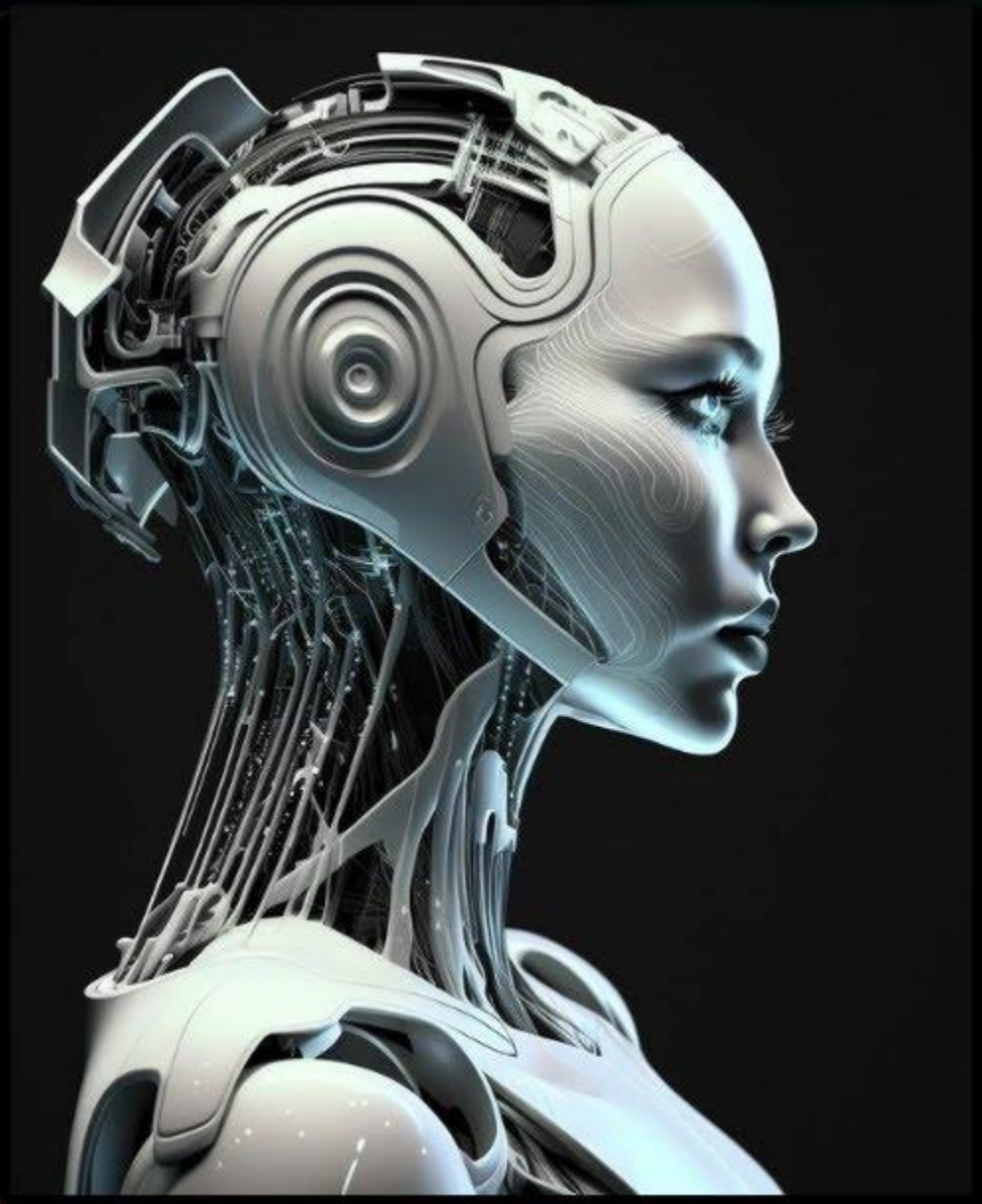
- February 2023
- 7, 13, 33 and 65 billion parameters.

LLama

LLaMA 2

- July 2023
- 7, 13 and 70 billion parameters.
- Same Arch + 40% more data.

Alpaca	7B, 13B	52k GPT-3 instructions
Vicuna	7B, 13B	70k ChatGPT conversations
Koala-distill	7B, 13B	117k cleaned ChatGPT conversations
GPT4-x-Alpaca	13B	20k GPT4 instructions
WizardML	7B	70k instructions synthesized with ChatGPT/GPT-3
OpenAssistant LLaMA	13B, 30B	600k human interactions (OpenAssistant Conversations)
More ..		



INTRODUCING

OpenAI Microsoft

GPT-4

March 14, 2023

MISTRAL AI_

September 27, 2023

HuggingFaceH4 / **zephyr-7b-beta** like 1.1k

Text Generation | Transformers | PyTorch | Safetensors | HuggingFaceH4/ultrachat_200k | HuggingFaceH4/ultrafeedback_binarized

English | mistral | generated_from_trainer | Eval Results | Inference Endpoints | text-generation-inference | arxiv:2305.18290

arxiv:2310.16944 | License: mit

Model card | Files | Community 42 | Train | Deploy | Use in Transformers

Edit model card

Downloads last month
174,280

Safetensors | Model size | 7.24B params

Tensor type | BF16

Model Card for Zephyr 7B β

Inference API

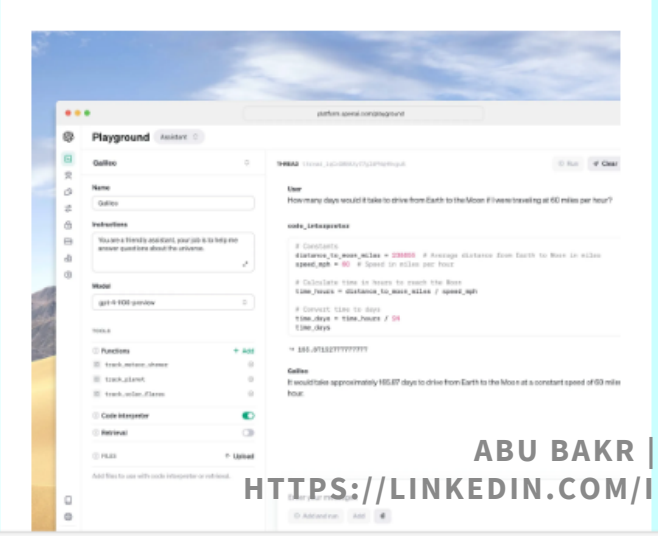
Recap

OpenAI's first developer conference

November 6, 2023 · San Francisco, CA

Thank you for joining us on OpenAI DevDay, OpenAI's first developer conference. Recordings of the keynote and breakout sessions will be available one week after the event.

In the meantime, learn more about OpenAI DevDay announcements for new models, developer products, and ChatGPT on our blog for highlights of the day.



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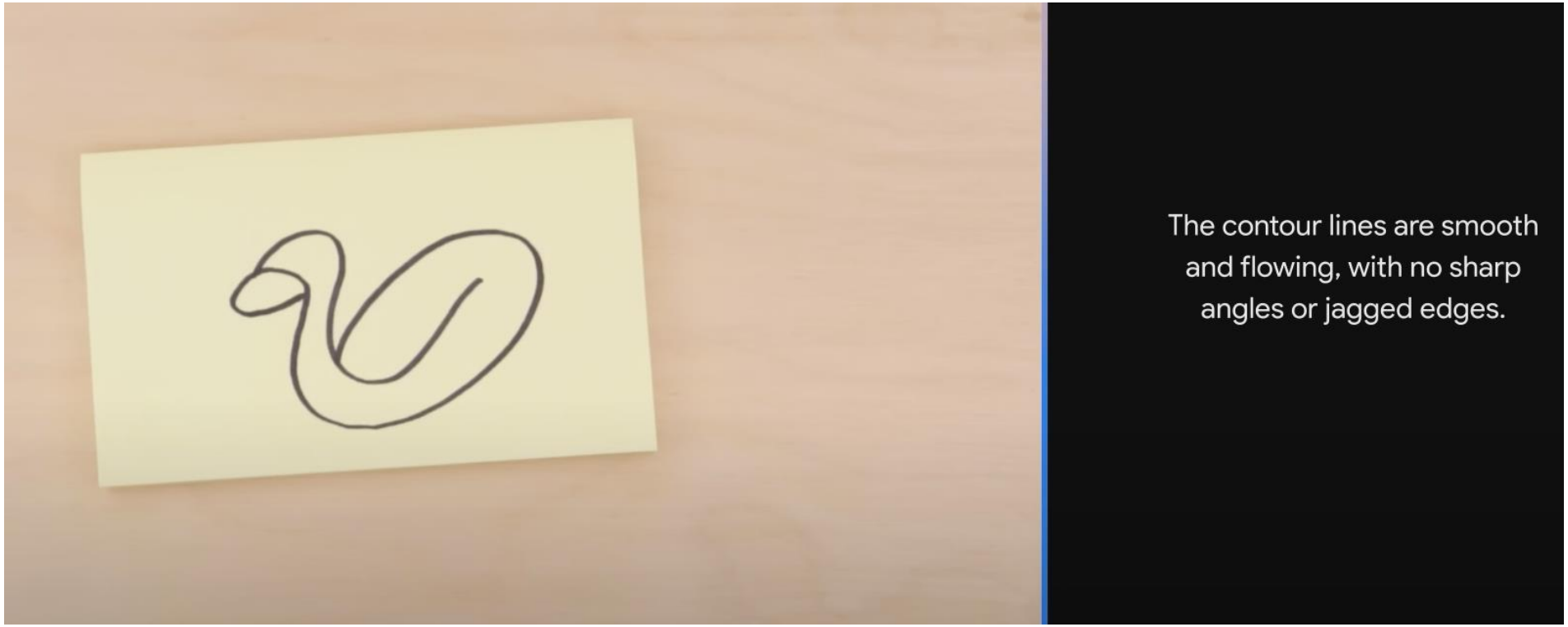
December 6, 2023

I can read, see and talk



I can read, see and talk

We've been testing the capabilities of Gemini, our new multimodal AI model.

We've been capturing footage to test it on a wide range of challenges, showing it a series of images, and asking it to reason about what it sees.

MoE, again

MISTRAL AI_

	LLaMA 2 70B	GPT - 3.5	Mixtral 8x7B
MMLU (MCQ in 57 subjects)	69.9%	70.0%	**70.6%**
HellaSwag (10-shot)	87.1%	85.5%	86.7%
ARC Challenge (25-shot)	85.1%	85.2%	**85.8%**
WinoGrande (5-shot)	**83.2%**	81.6%	81.2%
MBPP (pass@1)	49.8%	52.2%	**60.7%**
GSM-8K (5-shot)	53.6%	57.1%	**58.4%**
MT Bench (for Instruct Models)	6.86	**8.32**	8.30

of experts

Sparse Mixture-of-Experts.

What’s Next?

Julien Chaumond • Following
CTO at Hugging Face
4h •

my prediction for 2024 (yes, i have only one) 💡

Local ML is going to be huge.

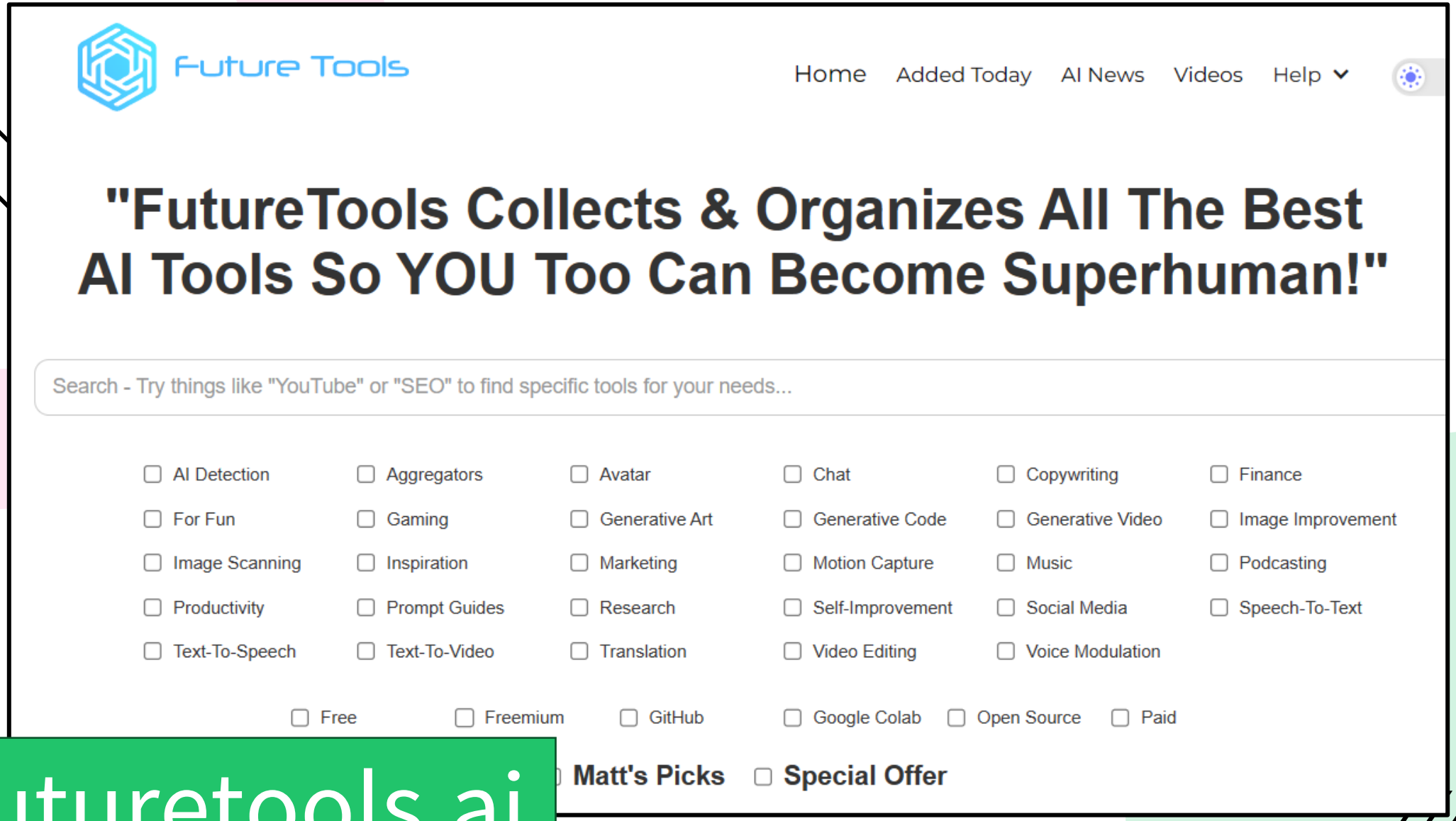
It will be in part driven by the adoption of Apple Silicon and other innovative hardware, but also on raw CPU and mobile devices

In many cases except for the largest of LLMs, local inference will become a viable alternative to hosted inference.

Valuates Reports

Large Language Model (LLM) Market

The Large Language Model (LLM) Market was valued at 10.5 Billion USD in 2022 and is anticipated to reach 40.8 Billi on USD by 2029, witnessing a CAGR of 21.4% during the forecast period 2023-2029.



Futuretools.ai

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